



Identifying Quantitative and Explanatory Tumor Indexes from Dynamic Contrast Enhanced Ultrasound

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Abstract. Contrast-enhanced ultrasound (CEUS) has been one of the most promising imaging techniques in tumor differential diagnosis since the real-time view of intra-tumor blood microcirculation. Existing studies primarily focus on extracting those discriminative imaging features whereas lack medical explanations. However, accurate quantitation of some clinical experience-driven indexes regarding intra-tumor vascularity, such as tumor infiltration and heterogeneity, still faces significant limitations. To tackle this problem, we present a novel scheme to identify quantitative and explanatory tumor indexes from dynamic CEUS sequences. Specifically, our method mainly comprises three steps: 1) extracting the stable pixel-level perfusion pattern from dynamic CEUS imaging using an improved stable principal component pursuit (SPCP) algorithm; 2) performing local perfusion variation comparison by the proposed Phase-constrained Wasserstein (PCW) distance; 3) estimating three clinical knowledge-induced tumor indexes, i.e. infiltration, regularity, and heterogeneity. The effectiveness of this method was evaluated on our collected CEUS dataset of thyroid nodules, and the resulting infiltration and heterogeneity index with $p < 0.05$ between different pathological types validated the efficacy of this quantitation scheme in thyroid nodule diagnosis.

Keywords: Contrast enhanced ultrasound · Perfusion analysis · Quantitative parameters estimation

1 Introduction

As a ‘hallmark of cancer’, tumor-induced angiogenesis acts as one of the most important biological behaviors by which tumors adapt to the change of nutrient requirement. Since the use of micro-bubble contrast agent, contrast-enhanced

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M. de Bruijne et al. (Eds.): MICCAI 2021, LNCS 12908, pp. 638–647, 2021.

https://doi.org/10.1007/978-3-030-87237-3_61

ultrasound (CEUS) provides a real-time visualization of dynamic tissue perfusion around lesion regions, and thus widely applied in differential diagnosis of various types of tumors [1].

Computer-aided analysis of dynamic CEUS imaging is quite challenging due to the following three aspects: Firstly, owing to irregular microbubble destruction during the acquisition of CEUS imaging, the inconsistency of intensity changes at each pixel is inevitable among adjacent temporal points; Secondly, there exists local temporal distortions when comparing pixel-wise perfusion patterns; Thirdly, existing tumor characteristics concluded by clinical experiences are especially qualitative and subjective, lack of precise numerical values for tumor quantitation [2].

Recently, existing studies mainly concentrate on the automated extraction of image features with the ‘Black Box’ characteristic. For example, traditional quantitative assessments of CEUS imaging rely on regional averaged Time Intensity Curves (TICs) by extracting a number of kinetics parameters. However, these features have no direct link to biological behaviors over tumor progression and lose spatial information. In addition, most machine learning or deep learning-based methods attempt to extract high-dimensional imaging features and learn latent representations, which is hard to be interpreted from a clinical point [3].

Different from previous studies, our study aims at quantifying some clinical experience-driven indexes to aid clinical tumor diagnosis. From the clinical point of view, professional radiologists concluded a few experienced characteristics of dynamic CEUS imaging that are commonly used in daily practices. These non-quantifiable indexes mainly include, lesion infiltration [4], regularity and homogeneity [5]. To the best of our knowledge, quantifying these experience-driven and explanatory tumor indexes from dynamic CEUS imaging has not yet been well studied.

To address these issues, we proposed a novel scheme to identify these quantitative and explanatory tumor indexes from CEUS imaging. The main innovations of this scheme include: 1) an improved Stable Principal Component Pursuit (SPCP) technique is adopted to extract stable pixel-wise perfusion patterns; 2) a novel Phase-constrained Wasserstein Distance (PCW) is proposed for pixel-level perfusion patterns analysis; 3) clinical experience-driven indexes of tumors are quantified based on the derived Area under the curve (AUC) and PCW view.

2 Method

Figure 1 presents a graphical overview of the proposed scheme. Our method consists of three main steps: 1) stable pixel-wise perfusion pattern extraction by the Stable Principal Component Pursuit (SPCP) (Sect. 2.1); 2) pixel-wise perfusion pattern analysis using Phase-constrained Wasserstein Distance (PCW) (Sect. 2.2); and 3) explanatory tumor indexes quantification based on the Area under the curve (AUC) and PCW views (Sect. 2.3).

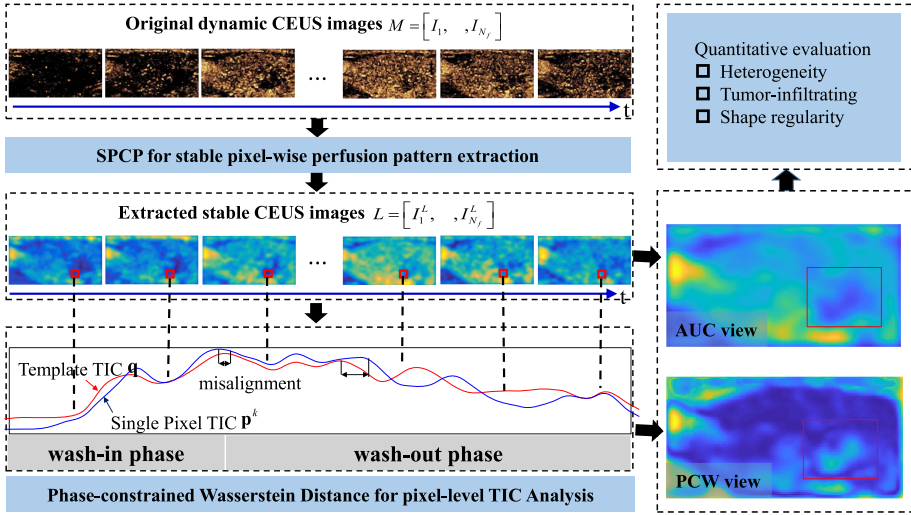


Fig. 1. Graphical overview of the proposed scheme. The input is sequential dynamic CEUS frames M and the output is the quantified tumor indexes which simultaneously preserve the spatial information and have clinical explanation. The overall quantitation process consists of three stages, 1) stable pixel-wise perfusion pattern extraction by the Stable Principal Component Pursuit (SPCP); 2) pixel-wise TIC analysis using Phase-constrained Wasserstein Distance (PCW); 3) explanatory tumor indexes quantification based on the Area under the curve (AUC) and PCW views.

2.1 SPCP for Stable Pixel-Wise Perfusion Pattern Extraction

Pixel-Wise Perfusion Pattern. Given one CEUS sequence $M = [I_1, \dots, I_{N_f}]$, where I_t is the vectorization of the t_{th} 2D CEUS frame and N_f is the length of the whole perfusion sequence. The k_{th} row of CEUS matrix $\mathbf{p}^k = [p_1^k, \dots, p_{N_f}^k]$ denotes the Time Intensity Curve (TIC) at the k_{th} pixel ($k = 1, \dots, N_j$, N_j is the number of pixels in each frame), also called pixel-wise perfusion pattern in this work.

Stable Pixel-Wise Perfusion Pattern Extraction. Although pixel-level TICs provide us an opportunity to evaluate perfusion pattern at the maximal spatial resolution compared with regional TICs, irregular microbubble oscillation or destruction might lead to inconsistent intensity fluctuation among adjacent frames at the pixel level. That is, apparent intensity change is a mixture of inherent microbubble concentration variations caused by intra-tumor blood supply and irregular fluctuations incurred by microbubble disruption. Therefore, an accurate evaluation of pixel-wise perfusion patterns depends on an effective recovery of fluctuation-free pixel-level TICs.

To extract the stable pixel-wise perfusion pattern, we attempt to decompose the original CEUS sequence M into a low rank component $L = [I_1^L, \dots, I_{N_f}^L]$

(i.e. the slow and smoothing intensity changes) and a sparse component $S = [I_1^S, \dots, I_{N_f}^S]$ (i.e. the rapid and irregular intensity fluctuations). In addition, for general CEUS imaging, an additional noise component ($\varepsilon > 0$) also needs to be considered. To accomplish this matrix decomposition, we apply an improved Stable Principal Component Pursuit (SPCP) algorithm [7], which is formulated as,

$$\begin{aligned} \min_{L, S} \max (\|L\|_*, \alpha \|S\|_1) \\ \text{s.t. } \|L + S - M\|_2 \leq \varepsilon \end{aligned} \quad (1)$$

where $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_*$ denote l_1 -norm, l_2 -norm, and nuclear norm respectively, α is the trade-off parameter which is fine-tuned according to the coefficient of determination R^2 in [6]. Equation(1) is a typical convex optimization problem which is solved by quasi-Newton method [7]. Accordingly, each row of the solution L is regarded as the stable perfusion pattern at one pixel position.

2.2 Phase-Constrained Wasserstein Distance for Perfusion Pattern Analysis

Template Perfusion Pattern. To refine tumor edge extraction and quantify internal heterogeneity, we resort to find a template perfusion pattern $\mathbf{q} = [q_1, \dots, q_{N_f}]$, which satisfies that distances between $\mathbf{q}$ and $\mathbf{p}_b^k$ (pixel-wise perfusion patterns from boundary) are nearly to zero as possible, while distances between $\mathbf{q}$ and $\mathbf{p}_t^k$ (those from intratumor) are large enough to quantify the local perfusion variations. Hence, we utilize the average of TICs from the coarse boundary outlined manually as the template $\mathbf{q}$.

Difference of Pixel-Level Perfusion Pattern. The next step is to define an effective distance metric for pixel-wise perfusion pattern comparison, such that local blood supply variation could be highlighted. Ideally, the resulting parametric map based on the new distances between $\mathbf{p}^k$ and $\mathbf{q}$ is able to delineate tumor boundary and reveal internal vascularity distinction. As shown in Fig. 1, local temporal alignment is essential for enhancement degree (i.e., blood supply) comparison since there exist temporal misalignments between pixel-wise TIC, i.e., instances of different sequences in the same temporal position may correspond to different enhancement phases.

Phase-Constrained Wasserstein Distance. Wasserstein distance provides a flexible way to match instances of two sets based on their pair-wise similarity [8]. As for dynamic CEUS sequence comparison, we propose a novel Phase-constrained Wasserstein distance (PCW), where a phase-related temporal regularization is added to ensure that matched intensities are temporally adjacent. Additionally, clinical diagnosis experiences tell us that the phase of contrast agent uptake generally contains more biological behavior information than that of wash-out phase, which suggests that different phases should have different matching flexibilities.

Given $\mathbf{p}^k = [p_1^k, \dots, p_{N_f}^k]$ and $\mathbf{q} = [q_1, \dots, q_{N_f}]$, by viewing intensity p_t^k and q_t as samples drawn from discrete distributions f and g respectively, Wasserstein distance is taken as the metric between TICs. The objective of Wasserstein distance is to find an optimal coupling matrix T (which belongs to $U = \{T \in R_+^{N_f \times N_f} | T1_{N_f} = f, T^T 1_{N_f} = g\}$) that transport the distribution f to g with the minimal cost. For simplicity, f and g are both viewed as uniform discrete distribution, and $d_{ij} = (p_i^k - q_j)^2$ is the squared distance between p_i^k and q_j . Thus, Wasserstein distance between TICs is formulated as,

$$d_{wd}(\mathbf{p}^k, \mathbf{q}) = \sum_{i=1}^{N_f} \sum_{j=1}^{N_f} T_{ij} d_{ij} \quad (2)$$

To encourage temporally adjacent elements to be matched, we introduce the phase-constrained temporal regularization $w(T)$, where $w(T) = KL(T||Q) = \sum_{i=1}^N \sum_{j=1}^N T_{ij} \log \frac{T_{ij}}{Q_{ij}}$ is the Kullback-Leibler divergence between two coupling matrices. The matrix Q represents the prior distribution of matching weights in T ,

$$Q_{ij} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(i-j)^2}{2\sigma^2}} \quad (3)$$

where σ controls the size of nonzero area around the diagonal of Q , and higher value indicates larger local matching flexibility. In order to control matching flexibility for different phases, we partition Q into two submatrices Q_{in} and Q_{out} , $Q = \begin{bmatrix} Q_{in} & \mathbf{0} \\ \mathbf{0} & Q_{out} \end{bmatrix}$, where $Q_{in} \in R_+^{N_p \times N_p}$, $Q_{out} \in R_+^{(N_f - N_p) \times (N_f - N_p)}$. N_p denotes Time to Peak (TTP) of template TIC, and interval $[0, N_p]$ and $[N_p, N_f]$ represent the wash-in and wash-out phase, respectively. In this work, we set $\sigma_{in} < \sigma_{out}$ to ensure temporal alignments stricter in wash-in phase. Hence, the proposed Phase-constrained Wasserstein Distance (PCW) is formulated as,

$$PCW(\mathbf{p}^k, \mathbf{q}) = \min_{T \in U(f, g)} \sum_{i=1}^{N_f} \sum_{j=1}^{N_f} T_{ij} d_{ij} + \lambda KL(T || Q) \quad (4)$$

where λ is the trade-off parameter. The Lagrangian function $L(T, f, g)$ is written as $L(T, \theta, v) = \sum_{i=1}^N \sum_{j=1}^N T_{ij} d_{ij} + \lambda T_{ij} \log \frac{T_{ij}}{Q_{ij}} + \theta^T (T1_N - f) + v^T (T^T 1_N - g)$. By setting the derivative $\frac{\partial L}{\partial T_{ij}}$ to zero, we can obtain $T^\lambda = e^{diag(-\frac{1}{2} - \frac{\theta}{\lambda})} K e^{diag(-\frac{1}{2} - \frac{v}{\lambda})}$, $k_{ij} = e^{-\frac{d'_{ij}}{\lambda}}$, $d'_{ij} = d_{ij} + \frac{\lambda}{2} \frac{(i-j)^2}{\sigma_{in(out)}^2}$. Motivated by the Sinkhorn distance, the solution of Eq. 4 can be efficiently obtained by matrix scaling algorithm [9].

2.3 PCW and AUC View for Quantifying the Explanatory Tumor Indexes

Tumor Infiltration. First we define the outer-edge of tumor $\mathbf{s}(i), i = 1, 2 \dots n_1$, a slightly expanded contour compared with original edge $\mathbf{c}(j), j = 1, 2 \dots n_2$, $n_1 \geq n_2$, and then the degree of tumor infiltration is defined as the Dynamic time warping (DTW) divergence of sequences based on AUC view [10], $TI = dtw(\mathbf{s}, \mathbf{c})$. The smaller TI means more similar vascular properties around the tumor edge, that is, more evident tumor invasion.

Shape Regularity. Fractal dimension (FD) is widely used to quantify the complexity of geometric forms [11]. In our work, we take the ratio of FD to tumor size S as the degree of shape regularity (SR), $SR = \frac{1}{S} \frac{\log(N_r)}{\log(1/r)}$, where N_r is the number of self-similar shapes and r is the corresponding scaling factor. Apparently, higher SR value represents less shape regularity.

Perfusion Heterogeneity. We utilize the image entropy of tumor region based on AUC or PCW view to measure internal perfusion heterogeneity [12]. Similarly, the value of entropy is normalized by the size of tumor S , $PH = -\frac{1}{S} \sum_i w_i \log w_i$, where w_i is the statistical frequency of tumor area. By the definition, larger PH value indicates that vascularity distributions within tumor region are more heterogeneous.

3 Experimental Setup and Evaluation

Dataset. Our dataset is a collection of 55 consecutive patients (Gender, F/M = 35/20; Age, 45.8 ± 12.3 years) attending Affiliated Drum Tower Hospital, Medical School of Nanjing University for thyroid nodules examination from September 2016 to July 2018. Only solitary nodules are included in this study. Three pathological types of nodules were incorporated, i.e., 17 Nodular Goiter, 19 Adenoma, and 19 Papillary Thyroid Carcinoma. The average maximum diameter of nodules is 19.5 ± 7.1 mm. All patients were examined by an expert radiologist with over 10-year clinical experience. Examinations were performed on a Logiq E9 ultrasound scanner (GE Healthcare, Milwaukee, WI, USA) with the second-generation microbubble contrast SonoVue at a low mechanical index (0.05–0.08). Pathologies results of all cases were confirmed by biopsy or surgical specimens. Each video lasts around 3 min with a frame rate of 15 fps and the spatial resolution of dual-view imaging is 600×800 pixels, where conventional gray-level view and contrast-specific view are displayed side by side. Approval was obtained by the ethics review board of local hospital and the informed consent was obtained from patients before this study.

Experimental Setup. In order to obtain a reliable quantitative analysis results, standard pre-processing steps were performed on raw video data, including motion correction, co-registration, and speckle noise reduction [13], etc. In our experiments, part of the beginning frames (the uptake of contrast agent has not yet started) and ending frames (the concentration of contrast agents no longer declines) were removed from raw videos. For pixel-wise perfusion pattern extraction using Stable Principal Component Pursuit, the balancing parameter α was set around $\frac{1}{\sqrt{m \times n}}$ (where m and n are the width and height of CEUS view). The optimal α value was chosen based on the coefficient of determination R^2 in [6]. ε was fixed to 0.05 for noise component. For Phase-constrained Wasserstein Distance, σ_{in} and σ_{out} control the expected bandwidth of warping for wash-in and wash-out phase, respectively. We set σ_{in} and σ_{out} to 0.75 and 1.5 in this study, which were fine-tuned ranging from 0.25 to 5 with the gap of 0.25. The optimal values are determined when the mean distance between $\mathbf{q}$ and $\mathbf{p}_b^k$ is nearly to zero as possible. As for temporal regularization, we chose $\lambda = 5$ to balance the matching flexibility and the temporal constraint.

3.1 Evaluation of the Quantified Tumor Indexes

To evaluate the validity of our proposed quantitative clinical experience-driven indexes in tumor differential diagnosis, we compared tumor infiltration(*TI*), shape regularity(*SR*) and perfusion heterogeneity(*PH*) from the perspective of benign and malignancy nodules and three pathological subtypes.

Table 1. Quantitative parameters. Except for *SR*, the other two indexes *TI* and *PH* (heterogeneity based on AUC and PCW parametric map respectively) show significant difference between benign and malignant nodules ($p \leq 0.05$)

Indexes	Benign	Malignant	p-value
Infiltration (TI)	3.09 ± 1.59	2.41 ± 0.81	0.04
Shape regularity (SR)	1.18 ± 0.73	1.33 ± 0.78	>0.05
Heterogeneity (PH from AUC)	3.84 ± 2.03	5.95 ± 3.68	0.03
Heterogeneity (PH from PCW)	3.56 ± 1.79	6.46 ± 4.21	0.01

First, Student t-test was performed to check for a significant difference between benign and malignant nodules, and the results are listed in Table 1. As can be observed, heterogeneity derived from PCW parametric map shows stronger discrimination than that estimated from AUC view, which confirms that PCW has the ability to characterize the pixel-level perfusion difference, leading to an effective perfusion index for differentiating benign and malignant nodules. Second, we further assess the quantitative indexes among three pathological types, Fig. 2 shows the boxplot of the three parameters, among which, degree of perfusion heterogeneity is calculated both on AUC and PCW maps.

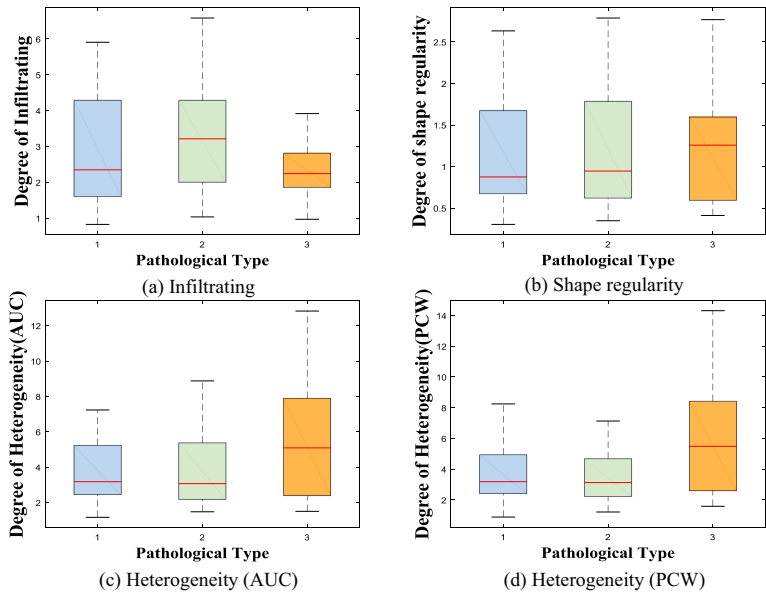


Fig. 2. Boxplots for values of three quantitative indexes among 1-Nodular Goiter, 2-Thyroid Adenoma, and 3-Papillary Thyroid Carcinoma.

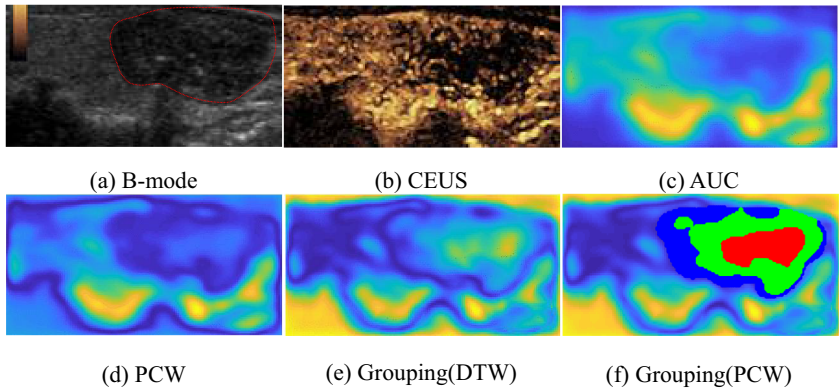


Fig. 3. A visualization of CEUS data and parametric maps, red line marks the manually outlined boundary. (a) B-mode gray-scale image I_{gs} ; (b) Contrast-enhanced imaging I_t ; (c-e) Parametric maps based on various estimated parameters; (f) Grouping result based on PCW view;

As for TI , the value of Papillary Thyroid Carcinoma is significantly smaller than Nodular Goiter and Thyroid Adenoma. Regarding benign nodules, Nodular Goiter has slightly stronger invasion than Thyroid Adenoma. Comparatively, shape of regularity is less discriminative among different pathological types.

Finally, Fig. 3 provides a visualization of tumor infiltration, shape regularity, heterogeneity and grouping results with the same spatial resolution as original CEUS. Apparently, PCW based map improved the enhancement along the boundary of nodule, meanwhile, in terms of internal heterogeneity, more abundant details on hemodynamic distribution of nodules is marked in PCW view than that of AUC and DTW maps.

4 Conclusion

We presented a novel scheme for the quantitative evaluation of dynamic CEUS imaging. In this framework, we first apply stable principal component pursuit to extract stable and smoothing perfusion pattern at each pixel, then utilize the proposed Phase-constrained Wasserstein distance to refine tumor delineation extraction and derive the parametric map (i.e., AUC and PCW view) with the same spatial resolution, and finally introduces three clinical experience-induced indexes for quantitative perfusion analysis. Currently, template TIC is determined by pixel-wise TICs from initial manually outlined boundary, which might degrade the eventual quantitation performance. It could be a promising direction to unify tumor boundary and internal heterogeneity information into template TIC selection in the future work.

Acknowledgement. This work was supported by the National Natural Science Foundation of China (Nos. 62136004, 61876082, 61732006, 61901214, U20A20389), the General Research Fund from Hong Kong Research Grants Council (Nos. 15205919), the National Key R&D Program of China (Grant Nos. 2018YFC2001600, 2018YFC2001602), the Nanjing Medical Science and Technique Development Foundation (Nos. YKK19054), and also by the CAAI-Huawei MindSpore Open Fund.

References

1. Jiang, Y., Zhou, Q., Zhu, C.: Contrast-enhanced ultrasound in the diagnosis of benign and malignant hepatic tumors. *J. Cent. South Univ. Med. Sci.* **37**(1), 53–60 (2012)
2. Balleyguier, C., et al.: New potential and applications of contrast-enhanced ultrasound of the breast: Own investigations and review of the literature. *Eur. J. Radiol.* **69**(1), 14–23 (2009)
3. Moon, W.K., Chen, I.L., Chang, J., Shin, S., Lo, C., Chang, R.: The adaptive computer-aided diagnosis system based on tumor sizes for the classification of breast tumors detected at screening ultrasound. *Ultrasonics* **76**, 70–77 (2017)
4. Wang, Y.M., Fan, W., Zhao, S., Zhang, K., Zhang, L., Zhang, P., et al.: Qualitative quantitative and combination score systems in differential diagnosis of breast lesions by contrast-enhanced ultrasound. *Eur. J. Radiol.* **85**(1), 48–54 (2016)
5. Hocke, M., Schulze, E., Gottschalk, P., Topalidis, T., Dietrich, C.F.: Contrast-enhanced endoscopic ultrasound in discrimination between focal pancreatitis and pancreatic cancer. *World J. Gastroenterol.* **12**(2), 246–50 (2006)

6. Christofides, D., Leen, E., Averkiou, M.A.: Automatic respiratory gating for contrast ultrasound evaluation of liver lesions. *IEEE Trans. Ultrason. Ferroelectr. Freq. Control* **61**(1), 25–32 (2014)
7. Aravkin, A., Becker, S., Cevher, V., Olsen, P.: A variational approach to stable principal component pursuit. In: *Proceedings of the Thirtieth Conference on Uncertainty in Artificial Intelligence, UAI*, vol. 2014, pp. 32–41 (2014)
8. Bing, S., Hua, G.: Order-preserving Wasserstein distance for sequence matching. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, CVPR*, vol. 2017, pp. 1049–1057 (2017)
9. Cuturi, M.: Sinkhorn distances: lightspeed computation of optimal transport. In: *Proceedings of the 27th Annual Conference on Neural Information Processing Systems, NIPS 2013*, pp. 2292–2300 (2013)
10. Keogh, E.J., Ratanamahatana, C.: Exact indexing of dynamic time warping. *Knowl. Inf. Syst.* **7**(3), 358–386 (2003)
11. Balankin, A.S., Ochoa, D.S., Miguel, I.A., Ortiz, J.: Fractal topology of hand-crumpled paper. *Phys. Rev. E* **81**(6), 11–26 (2010)
12. Yanai, K., Barnard, K.: Image region entropy: a measure of visualness of web images associated with one concept. In: *Proceedings of the 13th ACM International Conference on Multimedia, ACM 2005*, pp. 419–422 (2005)
13. Coupé, P., Hellier, P., Kervrann, C., Barillot, C.: Nonlocal means-based speckle filtering for ultrasound images. *IEEE Trans. Image Process.* **18**(10), 2221–2229 (2009)